
AWS VPC Flow Logs – Complete Guide + Lab

What Are VPC Flow Logs?

VPC Flow Logs capture IP traffic metadata going to and from network interfaces in your VPC.

Think of it as a **"Wireshark-lite" for AWS networking**, giving you visibility into **who's talking to what**, when, and how.

Key Use Cases

Use Case	Description
 Security monitoring	Detect suspicious access or communication attempts
 Troubleshooting	Analyze connectivity issues between EC2, RDS, etc.
 Traffic analytics	See traffic volume, patterns, and flow characteristics
 Compliance & auditing	Record and retain traffic history for audit purposes

Where Can You Enable Flow Logs?

You can enable logs at 3 levels:

- VPC

- **Subnet**
- **Network Interface (ENI)**

They all behave similarly, just different scope.



What Data Do Flow Logs Capture?

Each flow log record includes:

version account-id interface-id srcaddr dstaddr srcport dstport protocol packets bytes start end
action log-status

Example:

```
2 123456789012 eni-abc123 10.0.1.5 172.31.0.10 443 49252 6 3 150 1627833600 1627833660  
ACCEPT OK
```

Field	Description
srcaddr	Source IP
dstaddr	Destination IP
srcport	Source port
dstport	Destination port
protocol	Protocol number (6 = TCP, 17 = UDP)
packets	Number of packets
bytes	Bytes transferred
action	ACCEPT or REJECT



Important Notes

- VPC Flow Logs **do not capture** traffic:

- To/from **Amazon DNS**
 - Through **VPC endpoints**
 - Between **instances via localhost**
 - From AWS services to managed resources
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Lab: Enable and Analyze VPC Flow Logs



Lab Setup

Goal:

- Enable flow logs for a VPC
 - Send traffic between EC2 instances
 - View logs in CloudWatch
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Step 1: VPC and EC2 Setup

If not already created:

1. Create VPC: **10.0.0.0/16**
2. Subnets:
 - Public: **10.0.1.0/24**
 - Private: **10.0.2.0/24**
3. Create two EC2 instances:
 - EC2-A (public subnet)

- EC2-B (private subnet)
 - Allow ICMP, SSH between them
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✓ Step 2: Create IAM Role for Flow Logs

1. Go to **IAM** → **Roles** → **Create role**
 - Use Case: **VPC Flow Logs**
 - Attach: **CloudWatchLogsFullAccess**
 - Name: **VPCFlowLogsRole**
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✓ Step 3: Create a CloudWatch Log Group

1. Go to **CloudWatch** → **Logs** → **Log groups**
 2. Create a log group: **/vpc/flowlogs**
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✓ Step 4: Enable VPC Flow Logs

1. Go to **VPC** → **Your VPCs** → **Actions** → **Create flow log**
2. Filter: **All** (or **Accept**, **Reject**)
3. Destination: **Send to CloudWatch Logs**
4. Log group: **/vpc/flowlogs**
5. IAM role: **VPCFlowLogsRole**
6. Click **Create Flow Log**

 Wait 2–3 minutes for logs to populate.

✓ **Step 5: Generate Some Traffic**

From EC2-A:

```
ping <EC2-B private IP>  
curl <EC2-B private IP>:80
```

From EC2-B:

```
ping <EC2-A private IP>
```

✓ **Step 6: View Flow Logs in CloudWatch**

1. Go to **CloudWatch** → **Log groups** → **/vpc/flowlogs**
2. Choose latest stream
3. Look for entries like:

```
10.0.1.10 10.0.2.20 443 49252 6 3 150 1627833600 ACCEPT OK
```

You can analyze:

- Source/destination IPs
 - Accepted vs rejected traffic
 - Port/protocol usage
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Cleanup

- Stop or terminate EC2 instances

- Delete flow logs
- Remove IAM role and log group (optional)

Summary

Feature	Description
Scope	VPC / Subnet / ENI
Destination	CloudWatch Logs or S3
Visibility	Network metadata (not packet contents)
Use Cases	Security, auditing, troubleshooting
Limitations	Doesn't capture DNS, VPC endpoints traffic
